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GROUNDING OF INTEGRATED EXTERNAL EMI SHIELD OF AN OPTICAL SENSOR MODULE

Abstract

Methods, systems, and apparatuses for grounding of integrated external electromagnetic interference (EMI) shields are provided. A portion of a module substrate, different from a circuit board substrate, may protrude past the module as a location for one or more grounding pads to which the EMI shield is grounded. The EMI shield may cover at least a portion of the thickness of the module substrate. The EMI shield may be comprised of one or more parts, which may be coupled to one another. The module substrate may include conductive vias. The EMI shield may include a sprayed and/or printed coating. The EMI shield may include a darkened coating.

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Background/Summary

TECHNOLOGICAL FIELD

[0001] Example embodiments of the present disclosure relate generally to grounding of integrated external electromagnetic interference (EMI) shields.

BACKGROUND

[0002] Electromagnetic interference shielding creates a Faraday cage effect which attenuates radiation of electromagnetic (EM) waves and/or prevents EM emissions from circuit components such as optical sensors from being emitted outside of the circuit board. Conductive metal cans, commonly fabricated through stamping and/or drawing sheets of metallic material, create a conductive envelopment around the host circuit board containing the optical sensor. These conventional manners of manufacturing, including circuit board level metal can shielding may introduce issues such as an increase in overall footprint of the shielded circuit board. Due to the manufacturing process of stamped metal sheet cans, there is a minimum radius of the corner of the cans, which increases the footprint of the EMI shield. Additionally, using multiple metal cans to cover irregularly shaped components can introduce points through which EM waves may leak, particularly at higher frequencies. Circuit board level metal can shielding, in some examples, does not prevent internal EMI crosstalk between components within the same EMI shield. Additionally, using circuit board level shielding incurs costs in terms of quality assurance to ensure proper assembly and grounding, the process of assembling the shields, and material costs for shielding elements.

[0003] The inventors have identified numerous areas of improvement in the existing technologies and processes, which are the subjects of embodiments described herein. Through applied effort, ingenuity, and innovation, many of these deficiencies, challenges, and problems have been solved by developing solutions that are included in embodiments of the present disclosure, some examples of which are described in detail herein.

BRIEF SUMMARY

[0004] Various embodiments described herein relate to an improved assembly and grounding of integrated external EMI shields (e.g., for optical sensor modules).

[0005] In accordance with some embodiments of the present disclosure, an example method of assembling and grounding integrated external EMI shields is provided. The method may comprise assembling a module comprising a module substrate, the module substrate protruding on at least one side of the module, wherein the protruding portion of the module substrate comprises one or more grounding pads; and coupling an EMI shield comprising at least one shield component to the module.

[0006] In some embodiments, the method may comprise applying an adhesive to the module for adhering the EMI shield to the module; and curing the adhesive.

[0007] In some embodiments, the module substrate protrudes on one side of the module, wherein the protruding portion of the module substrate comprises two or more discrete grounding pads.

[0008] In some embodiments, the EMI shield comprises discrete cutouts for preventing jetted solder from expanding beyond a desired thickness.

[0009] In some embodiments, the method may comprise applying jetted solder to the one or more grounding pads of the module substrate.

[0010] In some embodiments, the module substrate protrudes on one side of the module, wherein the protruding portion of the module substrate comprises one elongated grounding pad.

[0011] In some embodiments, the method may comprise applying conductive adhesive to the one elongated grounding pad to create a conductive connection between the module substrate and the EMI shield; and curing the conductive adhesive.

[0012] In some embodiments, the method may comprise applying solder paste to the one elongated grounding pad to create a conductive connection between the module substrate and the EMI shield; and reflowing the solder paste.

[0013] In some embodiments, the module substrate protrudes on all sides of the perimeter of the module, wherein the protruding portions of the module substrate comprise a perimeter grounding pad.
[0014] In some embodiments, the method may comprise applying conductive adhesive to the perimeter grounding pad to create a conductive connection between the module substrate and the EMI shield; and curing the conductive adhesive.
[0015] In some embodiments, the method may comprise applying solder paste to the perimeter grounding pad to create a conductive connection between the module substrate and the EMI shield; and reflowing the solder paste.
[0016] In some embodiments, the module substrate comprises one or more grounding pads on an underside of the module substrate.
[0017] In some embodiments, the method may comprise applying conductive adhesive to the one or more grounding pads on the underside of the module substrate; coupling a bottom EMI shield component to the module via the one or more grounding pads; curing the conductive adhesive; and coupling a top EMI shield component to the bottom EMI shield component.
[0018] In some embodiments, the coupling the top EMI shield component to the bottom EMI shield component comprises using mechanical coupling.
[0019] In some embodiments, the module substrate comprises conductive vias and wherein the conductive vias surround the perimeter of the module substrate.
[0020] In some embodiments, the module substrate comprises conductive vias and wherein the conductive vias are dispersed throughout the module substrate.
[0021] In some embodiments, the method may comprise applying masking to areas determined to remain free of coating; spraying the EMI shield with a conductive coating such that the conductive coating grounds the EMI shield to the one or more grounding pads; removing the masking; and curing the conductive coating.
[0022] In some embodiments, the method may comprise printing a conductive coating on the EMI shield; applying conductive adhesive to the one or more grounding pads; and curing the conductive adhesive.
[0023] In some embodiments, the method may comprise printing a conductive coating on the EMI shield; applying solder paste to the one or more grounding pads; and reflowing the solder paste.
[0024] In some embodiments, the method may comprise using a darkened EMI shield, wherein the darkened EMI shield is comprised of at least one of: nickel-phosphorous; oxidized nickel; graphene; or dark-painted material.
[0025] The above summary is provided merely for purposes of summarizing some example embodiments to provide a basic understanding of some aspects of the disclosure. Accordingly, it will be appreciated that the above-described embodiments are merely examples and should not be construed to narrow the scope or spirit of the disclosure in any way. It will also be appreciated that the scope of the disclosure encompasses many potential embodiments in addition to those here summarized, some of which will be further described below.

Description

BRIEF SUMMARY OF THE DRAWINGS
[0026] Having thus described certain example embodiments of the present disclosure in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
[0027] FIG. **1**A illustrates a perspective view of an exemplary EMI shield having cutouts in accordance with one or more embodiments of the present disclosure.
[0028] FIG. **1**B illustrates a head-on view of an exemplary EMI shield having cutouts in

accordance with one or more embodiments of the present disclosure.
[0029] FIG. **2** illustrates an exemplary method of assembling a module comprising an EMI shield having cutouts in accordance with one or more embodiments of the present disclosure.
[0030] FIG. **3**A illustrates a perspective view of an exemplary module comprising an EMI shield grounded using an elongated grounding pad in accordance with one or more embodiments of the present disclosure.
[0031] FIG. **3**B illustrates a cross-sectional view of an exemplary module comprising an EMI shield grounded using an elongated grounding pad in accordance with one or more embodiments of the present disclosure.
[0032] FIG. **4**A illustrates an exemplary method of assembling a module comprising an EMI shield grounded using an elongated grounding pad and conductive adhesive in accordance with one or more embodiments of the present disclosure.
[0033] FIG. **4**B illustrates an exemplary method of assembling a module comprising an EMI shield grounded using an elongated grounding pad and solder in accordance with one or more embodiments of the present disclosure.
[0034] FIG. **5**A illustrates an exemplary method of assembling a module comprising a protruding perimeter of the module substrate and corresponding perimeter grounding pads using solder in accordance with one or more embodiments of the present disclosure.
[0035] FIG. **5**B illustrates an exemplary method of assembling a module comprising a protruding perimeter of the module substrate and corresponding perimeter grounding pads using conductive adhesive in accordance with one or more embodiments of the present disclosure.
[0036] FIG. **6** illustrates an exemplary method of assembling a module comprising a multi-part EMI shield in accordance with one or more embodiments of the present disclosure.
[0037] FIG. **7**A illustrates a perspective view of an exemplary substrate comprising conductive vias surrounding its perimeter in accordance with one or more embodiments of the present disclosure.
[0038] FIG. **7**B illustrates a cross-sectional perspective view of an exemplary substrate comprising conductive vias connected to the ground plane which extends across the thickness of the substrate and surrounding the perimeter of the substrate in accordance with one or more embodiments of the present disclosure.
[0039] FIG. **8** illustrates an exemplary method of applying a conductive coating to a module comprising an elongated grounding pad in accordance with one or more embodiments of the present disclosure.
[0040] FIG. **9** illustrates an exemplary method of applying a conductive coating to a module comprising an elongated grounding pad in accordance with one or more embodiments of the present disclosure.
[0041] FIG. **10** illustrates a perspective view of an exemplary EMI shield comprising a coated cap component and an elongated grounding pad in accordance with one or more embodiments of the present disclosure.
[0042] FIG. **11** illustrates a perspective view of an exemplary EMI shield comprising a coated cap component and discrete grounding pads in accordance with one or more embodiments of the present disclosure.
[0043] FIG. **12** illustrates a perspective view of an exemplary EMI shield comprising at least a darkened portion in accordance with one or more embodiments of the present disclosure.
[0044] FIGS. **13**A-**13**K illustrate example flowcharts of operations for assembling modules with EMI shielding in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0045] Some embodiments of the present disclosure will now be described more fully herein with reference to the accompanying drawings, in which some, but not all, embodiments of the disclosure are shown. Indeed, various embodiments of the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout.

[0046] As used herein, the term “comprising” means including but not limited to and should be interpreted in the manner it is typically used in the patent context. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of.

[0047] The phrases “in various embodiments,” “in one embodiment,” “according to one embodiment,” “in some embodiments,” and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

[0048] The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

[0049] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments or it may be excluded.

Overview

[0050] Various embodiments of the present disclosure are directed to improved EMI shielding and grounding thereof. Various embodiments may include EMI shields, which may be comprised of metal cans, plastic cans, and/or other materials. Various embodiments may include EMI shielding for modules such as optical modules.

[0051] Various embodiments may include using a portion of the optical module substrate, different from the circuit board substrate, which protrudes past the optical module as a location for grounding pad(s) to which the EMI shield is grounded. Various embodiments may include the EMI shield (e.g., use of a metal can, injection moulding of plastic embedded with electrically conductive fillers, use of electrically conductive foam sheets, etc.) covering at least a portion of the thickness of the optical module substrate, which may attenuate EM radiation that may leak out through the substrate thickness itself. Various embodiments may include a multi-part (e.g., at least a two-part) metal can which couple to one another and are grounded to the to the underside of the optical module substrate. In various embodiments, the underside of the optical module substrate is the side opposite the side which comprises the optical module. Various embodiments may include a plurality of conductive vias within the perimeter of the optical module substrate, which may attenuate EM radiation that may leak out through the substrate thickness itself. Various embodiments may include an EMI shield comprising a sprayed and/or printed conductive coating, which may be directly applied to the module and/or to a can.

[0052] The present disclosure, in some examples, includes a number of advantages, including creating stronger solder joints via cutouts preventing solder from spreading beyond a desired thickness, attenuating EMI leakage from the substrate thickness itself via overhanging portions of EMI shields, grounding EMI shields for EMI mitigation, attenuating EMI leakage from the substrate thickness itself via plated vias dispersed throughout the module substrate (e.g., surrounding the perimeter of the module substrate), reducing optical crosstalk via using darkened and/or blackened coatings, and/or other advantages.

Exemplary Systems and Apparatuses

[0053] Embodiments of the present disclosure herein include systems and apparatuses for improvement of EMI shielding and grounding thereof which are described herein and may be implemented in various embodiments.

[0054] FIG. **1**A illustrates a perspective view of an exemplary EMI shield having cutouts in accordance with one or more embodiments of the present disclosure. FIG. **1**A shows a module **100**A, which may comprise EMI shield **101**, optical module substrate **102**, grounding pads **103***a* . . . N, and cutouts **104***a* . . . N. In various embodiments, the EMI shield **101** may be metal. In various embodiments, a protruding portion of the module substrate **102** may comprise the grounding pads **103***a* . . . N. In various embodiments, the grounding pads **103***a* . . . N are discrete grounding pads. For example, the discrete grounding pads may be "small" (e.g., relative to the EMI shield and the module as a whole). For example, the discrete grounding pad may range in size (e.g., preferably approximately two hundred microns in width and length). In various embodiments, the cutouts **104***a* . . . N are discrete cutouts. For example, the discrete cutouts may correspond in size to the discrete grounding pads.

[0055] FIG. **1**B illustrates a head-on view of an exemplary EMI shield having cutouts in accordance with one or more embodiments of the present disclosure. FIG. **1**B shows a module **100**B, which may comprise a head-on view of the cutouts **104***a* . . . N and solder **105***a* . . . N (e.g., solder balls). The solder **105***a* . . . N may be jetted solder and/or may be otherwise applied. In various embodiments, the cutouts **104***a* . . . N on the EMI shield **101** may allow for localized dispersion of the jetted solder **105***a* . . . N, for example, preventing the jetted solder from spreading beyond the confines of the cutouts **104***a* . . . N and becoming too thin, thereby mitigating potential solder crack issues. In various embodiments, the EMI shield **101** may extend down at least a portion of the thickness of the module substrate (e.g., overhanging the sides of the module substrate), which may allow the EMI shield to attenuate EMI leakage from the thickness of the substrate itself. For example, the EMI shield may extend down the thickness of the substrate on three of the four sides of the module substrate.

[0056] FIG. **2** illustrates an exemplary method **200** of assembling a module comprising an EMI shield having cutouts in accordance with one or more embodiments of the present disclosure. FIG. **2** illustrates the method **200** for assembling a module comprising an EMI shield having cutouts, such as those shown in FIGS. **1**A-**1**B. The method **200** shows the assembled module at **201**, the module with an attached adhesive (e.g., can glue) at **202**, attaching the EMI shield at **203**, curing the adhesive at **204**, solder jetting at **205***a*, and a blown-up view **205***b* of the solder jetting of **205***a*. In various embodiments, the cutouts **104***a* . . . N allow for a stronger solder joint by preventing the jetted solder **105***a* . . . N from expanding beyond a desired thickness.

[0057] FIG. **3**A illustrates a perspective view of an exemplary module comprising an EMI shield grounded using an elongated grounding pad in accordance with one or more embodiments of the present disclosure. FIG. **3**A shows a module **300**A, which may comprise an optical module **301**, an EMI shield **302**, adhesive **303**, and a grounding pad **304** (e.g., with adhesive). In various embodiments, the EMI shield **302** may extend down at least a portion of the thickness of the module substrate, which may allow the EMI shield to attenuate EMI leakage from the thickness of the substrate itself. For example, the EMI shield may extend down the thickness of the substrate on three of the four sides of the module substrate. The adhesive **303** may allow for attaching the EMI shield **302** to the optical module **301**.

[0058] FIG. **3**B illustrates a cross-sectional view of an exemplary module comprising an EMI shield grounded using an elongated grounding pad in accordance with one or more embodiments of the present disclosure. FIG. **3**B shows a module **300**B, which is a cross-sectional view of the module **300**A. The module **300**B shows the optical module **301**, the EMI shield **302**, the adhesive **303**, the grounding pad **304**, conductive adhesive **306** and a protruding portion of module substrate **305**. The protruding portion of the module substrate **305** may comprise at least one grounding pad (e.g., one elongated grounding pad). For example, the elongated grounding pad may be "large" relative to the size of the EMI shield and/or the module. For example, the elongated grounding pad may approximately correspond in size to the width of the EMI shield. For example, the conductive adhesive **306** (e.g., a silver-based adhesive, etc.) may be used to electrically couple the EMI shield **302** to the grounding pad **304**.

[0059] FIG. **4**A illustrates an exemplary method **400**A of assembling a module comprising an EMI shield grounded using an elongated grounding pad and conductive adhesive in accordance with one or more embodiments of the present disclosure. FIG. **4**A illustrates the method **400**A for assembling a module comprising an EMI shield and an elongated grounding pad with conductive adhesive, such as those shown in FIGS. **3**A-**3**B. The method **400**A shows an assembled module at **401**, dispensed adhesive for attaching the EMI shield **302** onto the top of the module **401** in provisioned glue pockets and dispensed conductive adhesive onto the protruded substrate grounding pad **304** at **402**, attaching the EMI shield at **403**, and curing the adhesives at **404**.

[0060] FIG. **4**B illustrates an exemplary method **400**B of assembling a module comprising an EMI shield grounded using an elongated grounding pad and solder in accordance with one or more embodiments of the present disclosure. FIG. **4**B illustrates the method **400**B for assembling a module comprising an EMI shield and an elongated grounding pad with solder, such as those shown in FIGS. **3**A-**3**B. The method **400**B shows an assembled module at **405**, dispense adhesive for attaching the EMI shield **302** onto the top of the module **401** in provisioned glue pockets and dispense solder paste onto the protruded substrate grounding pad **304** at **406**, attaching the EMI shield at **407**, curing the adhesive at **408**, and reflowing the dispensed solder paste at **409**.

[0061] FIG. **5**A illustrates an exemplary method **500**A of assembling a module comprising a protruding perimeter of the module substrate and corresponding perimeter grounding pads using solder paste in accordance with one or more embodiments of the present disclosure. The method **500**A shows an assembled module with a protruding perimeter of the module substrate and corresponding perimeter grounding pad(s) at **501**, dispensing solder paste on the grounding pad(s) at **502**, dispensing adhesive into pockets provisioned on the top of the module to attach the EMI shield on the module at **503**, attaching the EMI shield at **504**, curing the adhesive at **505**, and reflowing solder paste at **506**.

[0062] FIG. **5**B illustrates an exemplary method **500**B of assembling a module comprising a protruding perimeter of the module substrate and corresponding perimeter grounding pads using conductive adhesive in accordance with one or more embodiments of the present disclosure. The method **500**B shows an assembled module with a protruding perimeter of the module substrate and corresponding perimeter grounding pad(s) at **507**, dispensing conductive adhesive onto the perimeter grounding pads at **508**, dispensing adhesive into adhesive pockets on the top of the module for attaching the EMI shield to the module at **509**, attaching the EMI shield at **510**, and curing the adhesives at **511**.

[0063] FIG. **6** illustrates an exemplary method of assembling a module comprising a multi-part EMI shield in accordance with one or more embodiments of the present disclosure. The method **600** shows an assembled module at **601** with grounding pads **601***a* (e.g., at the corners of the underside of the substrate), dispensing conductive adhesive on the grounding pads **601***a* at **602**, coupling the bottom portion of EMI shield to module at **603**, curing the conductive adhesive at **604**, and coupling the top portion of EMI shield to the bottom portion at **605**. The multi-part EMI shield may be comprised of two or more parts. In various embodiments, the multi-part EMI shield may be comprised of two parts. In various embodiments, the first portion of the EMI shield (e.g., the bottom portion) may be placed on the underside of the assembled module. In various embodiments, the underside of the optical module substrate is the side opposite the side which comprises the optical module. In various embodiments, the second portion of the EMI shield (e.g., the top portion) may be placed over the top side of the module, which may form a complete enveloped EMI shielded module. In various embodiments, the top side of the module is the side which comprises the optical module. In various embodiments, the two or more portions of the EMI shield may be variously coupled to one another, for example, they may be mechanically coupled to one another. In various embodiments, to facilitate grounding of the EMI shield (e.g., a metal can), the optical module substrate may include grounding pads such as the grounding pads **601***a*, which may be variously located (e.g., on the underside of the module substrate). In various embodiments, conductive adhesive may be dispensed onto the grounding pads **601***a* before coupling the module to the first portion of the EMI shield. In various embodiments, the conductive adhesive may contact the grounding pads **601***a* of the module substrate to the EMI shield which may achieve grounding for EMI mitigation.

[0064] FIG. **7**A illustrates a perspective view of an exemplary substrate comprising conductive vias surrounding its perimeter in accordance with one or more embodiments of the present disclosure. The exemplary substrate **700**A shows a module substrate **701** and conductive vias **701***a* . . . N surrounding the perimeter of the module substrate **701**. In various embodiments, the perimeter vias **701***a* . . . N may be plated vias. In various embodiments, the perimeter vias **701***a* . . . N, when coupled to the ground of the module (e.g., grounding pads), may provide a conductive barrier against electromagnetic waves escaping through the thickness of the module substrate **701** itself. In various embodiments, perimeter vias may be combined with any other EMI shielding method to provide a complete EMI shield around the module, for example, such as the EMI shielding methods described herein (e.g., the EMI shielding methods of FIGS. **1**A-**6** and FIGS. **8-12**).

[0065] FIG. **7**B illustrates a cross-sectional perspective view of an exemplary substrate comprising conductive vias that are connected to the ground plane of the substrate and span the perimeter of the substrate. The exemplary substrate **700**B shows a module substrate **702** and conductive vias **702***a* . . . N which surround the perimeter of the module substrate **702**. In various embodiments, the vias **702***a* . . . N may be plated vias which extend across the entire thickness of the substrate and are electrically connected to the substrate ground. In various embodiments, the vias **702***a* . . . N, when coupled to the ground of the module (e.g., grounding pads), may provide a conductive barrier against electromagnetic waves escaping through the thickness of the module substrate **702** itself. In various embodiments, vias may be combined with any other EMI shielding method, for example, such as the EMI shielding methods described herein. In various embodiments, the vias **702***a* . . . N may be perimeter vias such as those described with respect to FIG. **7**A. In various embodiments, plated vias may be combined with any other EMI shielding method, for example, such as the EMI shielding methods described herein (e.g., the EMI shielding methods of FIGS. **1**A-**6** and FIGS. **8-12**).

[0066] FIG. **8** illustrates an exemplary method **800** of applying a conductive coating to a module comprising an elongated grounding pad in accordance with one or more embodiments of the present disclosure. The method **800** shows an example of spraying and/or printing a conductive coating (e.g., a polymer-based conductive coating) on a module for EMI shielding. The method **800** shows an assembled module with an elongated grounding pad on protruding portion of the module substrate at **801**, masking apertures at **802**, spraying and/or printing the conductive coating at **803**, and masking removal and curing of the conductive coating at **804**. In various embodiments, the conductive coating may be sprayed onto the module and/or onto a separate can. In various embodiments, the conductive coating may be printed onto the module and/or onto a separate can. In various embodiments, the spraying and/or printing of the conductive coating may also cover at least a portion of the thickness of the module substrate, which may allow for shielding against electromagnetic waves being emitted from the substrate thickness. In various embodiments, when printing the conductive coating, the printing may skip printing over the apertures/desired areas (e.g., instead of using masking over the apertures). In various embodiments, the sprayed and/or printed coating may be applied to the elongated grounding pad.

[0067] FIG. **9** illustrates an exemplary method **900** of applying a conductive coating to a module comprising an elongated grounding pad in accordance with one or more embodiments of the present disclosure. The method **900** shows a cap component (e.g., the cap component may be comprised of plastic) at **901**, coating the cap component (e.g., using electroless plating and/or physical vapor deposition (PVD)) with a conductive material (e.g., copper-nickel), assembling a module with the coated cap component at **903** (wherein the module substrate includes a protruding portion with an elongated grounding pad **903***a*), applying conductive adhesive and/or solder to grounding pad at **904** such that conductive adhesive and/or solder is also in contact with the coated cap **904***a*, and curing (e.g., oven curing) and/or reflowing to set the conductive adhesive and/or solder at **905**.

[0068] FIG. **10** illustrates a perspective view **1000** of an exemplary EMI shield comprising a coated cap component and an elongated grounding pad in accordance with one or more embodiments of the present disclosure. The perspective view **1000** shows an assembled module **1001** comprising a coated cap **1002** (e.g., the cap may be coated with a metallic material such as copper-nickel via electroless plating and/or PVD, wherein the coating process may involve the use of masks to mask optical openings on the module to ensure coating does not cover these openings hindering the performance of the optical module), and solder paste dispensed on the elongated grounding pad **1003** on a protruding portion of the module substrate. In various embodiments, when the module is reflowed, the solder paste will couple the coated cap to the grounding pad on the module substrate.

[0069] FIG. **11** illustrates a perspective view **1100** of an exemplary EMI shield comprising a coated cap component and discrete grounding pads in accordance with one or more embodiments of the present disclosure. The perspective view **1100** shows an assembled module **1101** comprising a coated cap **1102** (e.g., the cap may be coated with a metallic material such as copper-nickel via electroless plating and/or PVD), and jetted solder **1103***a* . . . N dispensed in balls on discrete grounding pads separated by solder mask on a protruding portion of the module substrate. In various embodiments, the discrete grounding pads may be small grounding pads separated by solder mask to prevent jetted solder balls from spreading too wide and/or thin over the region. For example, the small grounding pads may be approximately one hundred microns.

[0070] FIG. **12** illustrates a perspective view **1200** of an exemplary EMI shield comprising at least a darkened portion in accordance with one or more embodiments of the present disclosure. The perspective view **1200** shows an assembled module **1201** comprising a cap with a darken to surface **1202**, and solder **1203** on an elongated grounding pad on a protruding portion of module substrate. In various embodiments, the darkening may be achieved by painting at least a portion of a conductive coated cap **1002** a dark color (e.g., black), applying a secondary coating which has anti-reflective properties, and/or using darkened/blackened conductive coatings (e.g., oxidized nickel, nickel-phosphorous, graphene, etc.). In various embodiments, the darkened cap may reduce optical crosstalk. A darkened/blackened EMI shield may be used in combination with one or more of the embodiments described herein, which may include darkening the top sides of the assembled EMI shield **101**.

[0071] It should be readily appreciated that the embodiments of the systems and apparatuses, described herein may be configured in various additional and alternative manners in addition to those expressly described herein.

Exemplary Methods

[0072] FIGS. **13**A-**13**K illustrate example flowcharts of operations for assembling modules with EMI shielding in accordance with one or more embodiments of the present disclosure.

[0073] FIG. **13**A illustrates an example flowchart **1300**A of operations for assembling a module comprising an EMI shield having cutouts in accordance with one or more embodiments of the present disclosure.

[0074] At operation **1301**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.

[0075] At operation **1302**, assembling the module such that the module substrate protrudes on one side of the module may be performed. In various embodiments, the protruding portion of the module substrate may comprise two or more discrete grounding pads (e.g., the grounding pads **103***a* . . . N).

[0076] At operation **1303**, applying an adhesive to the module may be performed. In various embodiments, the adhesive may allow for adhering an EMI shield to the module. In various embodiments, curing the adhesive may be performed. In various embodiments, the curing may comprise oven curing.

[0077] At operation **1304**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield comprises at least one shield component. In various embodiments, the EMI shield comprises discrete cutouts (e.g., the cutouts **104***a* . . . N) for preventing jetted solder from expanding beyond a desired thickness.

[0078] At operation **1305**, applying jetted solder to the two or more discrete grounding pads of the module substrate may be performed. In various embodiments, the jetted solder may be solder balls (e.g., the solder balls **105***a* . . . N).

[0079] FIG. **13**B illustrates an example flowchart **1300**B of operations for assembling a module comprising an EMI shield grounded using an elongated grounding pad and conductive adhesive in accordance with one or more embodiments of the present disclosure.

[0080] At operation **1306**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.

[0081] At operation **1307**, assembling the module such that the module substrate protrudes on one side of the module may be performed. In various embodiments, the protruding portion of the module substrate (e.g., protruding portion **305**) may comprise one elongated grounding pad (e.g., the grounding pad **304**).

[0082] At operation **1308**, applying an adhesive to the module for adhering an EMI shield to the module may be performed. In various embodiments, curing the adhesive may be performed. In various embodiments, the curing may comprise oven curing.

[0083] At operation **1309**, applying conductive adhesive to the one elongated grounding pad to create a conductive connection between the module substrate and the EMI shield may be performed.

[0084] At operation **1310**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.

[0085] At operation **1311**, curing the conductive adhesive may be performed. In various embodiments, the curing may comprise oven curing.

[0086] FIG. **13**C illustrates an example flowchart **1300**C of operations for assembling a module comprising an EMI shield grounded using an elongated grounding pad and solder paste in accordance with one or more embodiments of the present disclosure.

[0087] At operation **1312**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.

[0088] At operation **1313**, assembling the module such that the module substrate protrudes on one side of the module may be performed. In various embodiments, the protruding portion of the module substrate (e.g., protruding portion **305**) may comprise one elongated grounding pad (e.g., the grounding pad **304**).

[0089] At operation **1314**, applying an adhesive to the module for adhering an EMI shield to the module may be performed.

[0090] At operation **1315**, applying solder paste to the one elongated grounding pad to create a conductive connection between the module substrate and the EMI shield may be performed.

[0091] At operation **1316**, coupling the EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.

[0092] At operation **1317**, reflowing the solder paste and curing the adhesive may be performed. In various embodiments, the curing may comprise oven curing.

[0093] FIG. **13**D illustrates an example flowchart **1300**D of operations for assembling a module comprising a protruding perimeter of the module substrate and corresponding perimeter grounding pads using conductive adhesive in accordance with one or more embodiments of the present

disclosure.
[0094] At operation **1318**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.
[0095] At operation **1319**, assembling the module such that the module substrate protrudes on all sides (e.g., all four sides) of the perimeter of the module is performed. In various embodiments, the protruding portions of the module substrate comprise a perimeter grounding pad.
[0096] At operation **1320**, applying an adhesive to the module for adhering an EMI shield to the module may be performed. In various embodiments, curing the adhesive may be performed. In various embodiments, the curing may comprise oven curing.
[0097] At operation **1321**, applying a conductive adhesive to the perimeter grounding pad to create a conductive connection between the module substrate and EMI shield may be performed.
[0098] At operation **1322**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.
[0099] At operation **1323**, curing the adhesive may be performed. In various embodiments, the curing may be oven curing.
[0100] In various embodiments, the flowchart **1300**D may describe a method such as the one shown in FIG. **5**B.
[0101] FIG. **13**E illustrates an example flowchart **1300**E of operations for assembling a module comprising a protruding perimeter of the module substrate and corresponding perimeter grounding pads using solder in accordance with one or more embodiments of the present disclosure.
[0102] At operation **1324**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.
[0103] At operation **1325**, assembling the module such that the module substrate protrudes on all sides (e.g., all four sides) of the perimeter of the module is performed. In various embodiments, the protruding portions of the module substrate comprise a perimeter grounding pad.
[0104] At operation **1326**, applying an adhesive to the module for adhering an EMI shield to the module may be performed.
[0105] At operation **1327**, applying solder (e.g., solder paste) to the perimeter grounding pad to create a conductive connection between the module substrate and the EMI shield may be performed.
[0106] At operation **1328**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.
[0107] At operation **1329**, reflowing the solder and curing the adhesive may be performed. In various embodiments, the curing may comprise oven curing.
[0108] FIG. **13**F illustrates an example flowchart **1300**F for assembling a module comprising a multi-part EMI shield in accordance with one or more embodiments of the present disclosure.
[0109] At operation **1330**, assembling a module comprising a module substrate may be performed.
[0110] At operation **1331**, assembling the module such that the module substrate comprises one or more grounding pads on the underside of the module substrate may be performed.
[0111] At operation **1332**, applying conductive adhesive to the one or more grounding pads on the underside of the module substrate may be performed.
[0112] At operation **1333**, coupling a bottom EMI shield component (e.g., the first EMI shield component of FIG. **6**) to the module via the one or more grounding pads may be performed.
[0113] At operation **1334** curing the conductive adhesive may be performed. In various embodiments, the curing may comprise oven curing.
[0114] At operation **1335**, coupling a top EMI shield component (e.g., the second EMI shield component of FIG. **6**) to the bottom EMI shield component may be performed. In various embodiments, the coupling the top EMI shield component to the bottom EMI shield component

may comprise mechanical coupling.

[0115] FIG. **13**G illustrates an example flowchart **1300**G for assembling a module with a substrate comprising conductive vias surrounding its perimeter in accordance with one or more embodiments of the present disclosure.

[0116] At operation **1336**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.

[0117] At operation **1337** assembling the module such that the module substrate comprises conductive vias extending through the thickness of the substrate may be performed. In various embodiments, the conductive vias may surround the perimeter of the module substrate (e.g., such as shown in FIG. **7**A).

[0118] At operation **1338**, applying an adhesive to the module for adhering an EMI shield to the module may be performed.

[0119] At operation **1339**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.

[0120] At operation **1340**, curing the adhesive may be performed. In various embodiments, the curing may comprise oven curing.

[0121] FIG. **13**H illustrates an example flowchart **1300**I for assembling a module by applying (e.g., spraying) a conductive coating to a module comprising an elongated grounding pad in accordance with one or more embodiments of the present disclosure.

[0122] At operation **1346**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate comprises one or more grounding pads.

[0123] At operation **1347**, applying masking to areas determined to remain free of coating may be performed. For example, masking may be applied to apertures of an optical module to prevent coating of the apertures.

[0124] At operation **1348**, spraying and EMI shield (e.g., the module itself, such as described with respect to FIG. **8**) with a conductive coating such that the conductive coating grounds the EMI shield to the one or more grounding pads may be performed. In various embodiments, the conductive coating may be sprayed onto a cap component (e.g., such as described with respect to FIG. **9**).

[0125] At operation **1349**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.

[0126] At operation **1350**, removing the masking may be performed.

[0127] At operation **1350**, curing the conductive coating may be performed.

[0128] FIG. **13**I illustrates an example flowchart **1300**J for assembling a module by applying (e.g., printing) a conductive coating to a module comprising an elongated grounding pad and using conductive adhesive in accordance with one or more embodiments of the present disclosure.

[0129] At operation **1352**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate comprises one or more grounding pads.

[0130] At operation **1353**, applying masking to areas determined to remain free of coating may be performed. For example, masking may be applied to apertures of an optical module to prevent coating of the apertures. In various embodiments, such as when printing the conductive coating, applying the masking may be omitted.

[0131] At operation **1354**, printing a conductive coating onto an EMI shield may be performed. In various embodiments, the conductive coating may be printed directly onto the module itself (e.g., such as described with respect to FIG. **8**). In various embodiments, the conductive coating may be printed onto a cap component (e.g., such as described with respect to FIG. **9**).

[0132] At operation **1355**, coupling an EMI shield to the module may be performed. In various

embodiments, the EMI shield may comprise at least one shield component.
[0133] At operation **1356**, applying conductive adhesive to the one or more grounding pads may be performed.
[0134] At operation **1357**, curing the conductive adhesive may be performed. In various embodiments, the curing may comprise oven curing.
[0135] FIG. **13**J illustrates an example flowchart **1300**K for assembling a module by applying (e.g., printing) a conductive coating to a module comprising an elongated grounding pad and using solder in accordance with one or more embodiments of the present disclosure.
[0136] At operation **1358**, assembling a module comprising a module substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate comprises one or more grounding pads.
[0137] At operation **1359**, applying masking to areas determined to remain free of coating may be performed. For example, masking may be applied to apertures of an optical module to prevent coating of the apertures. In various embodiments, such as when printing the conductive coating, applying the masking may be omitted.
[0138] At operation **1360**, printing a conductive coating onto an EMI shield may be performed. In various embodiments, the conductive coating may be printed directly onto the module itself (e.g., such as described with respect to FIG. **8**). In various embodiments, the conductive coating may be printed onto a cap component (e.g., such as described with respect to FIG. **9**).
[0139] At operation **1361**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component.
[0140] At operation **1362**, applying solder (e.g., solder paste) to the one or more grounding pads may be performed.
[0141] At operation **1363**, reflowing the solder may be performed.
[0142] FIG. **13**K illustrates an example flowchart **1300**L for assembling a module with a darkened EMI shield in accordance with one or more embodiments of the present disclosure.
[0143] At operation **1364**, assembling a module comprising a substrate may be performed. In various embodiments, the module substrate may protrude on at least one side of the module, wherein the protruding portion of the module substrate may comprise one or more grounding pads.
[0144] At operation **1365**, applying an adhesive to the module for adhering an EMI shield to the module may be performed.
[0145] At operation **1366**, coupling an EMI shield to the module may be performed. In various embodiments, the EMI shield may comprise at least one shield component. In various embodiments, the EMI shield may be darkened by comprising at least one of the following: nickel-phosphorous, oxidized nickel, graphene, and/or a dark-painted material.
[0146] At operation **1367**, curing the adhesive may be performed. In various embodiments, the curing may be oven curing.

CONCLUSION

[0147] Operations and/or functions of the present disclosure have been described herein, such as in flowcharts. As will be appreciated, computer program instructions may be loaded onto a computer or other programmable apparatus (e.g., hardware) to produce a machine, such that the resulting computer or other programmable apparatus implements the operations and/or functions described in the flowchart blocks herein. These computer program instructions may also be stored in a computer-readable memory that may direct a computer, processor, or other programmable apparatus to operate and/or function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture, the execution of which implements the operations and/or functions described in the flowchart blocks. The computer program instructions may also be loaded onto a computer, processor, or other programmable apparatus to cause a series of operations to be performed on the computer, processor, or other programmable apparatus to produce a computer-implemented process such that the instructions executed on the

computer, processor, or other programmable apparatus provide operations for implementing the functions and/or operations specified in the flowchart blocks. The flowchart blocks support combinations of means for performing the specified operations and/or functions and combinations of operations and/or functions for performing the specified operations and/or functions. It will be understood that one or more blocks of the flowcharts, and combinations of blocks in the flowcharts, can be implemented by special purpose hardware-based computer systems which perform the specified operations and/or functions, or combinations of special purpose hardware with computer instructions.
[0148] While this specification contains many specific embodiments and implementation details, these should not be construed as limitations on the scope of any disclosures or of what may be claimed, but rather as descriptions of features specific to particular embodiments of particular disclosures. Certain features that are described herein in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.
[0149] While operations and/or functions are illustrated in the drawings in a particular order, this should not be understood as requiring that such operations and/or functions be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, operations and/or functions in alternative ordering may be advantageous. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results. Thus, while particular embodiments of the subject matter have been described, other embodiments are within the scope of the following claims.
[0150] While this detailed description has set forth some embodiments of the present invention, the appended claims cover other embodiments of the present invention which differ from the described embodiments according to various modifications and improvements.
[0151] Within the appended claims, unless the specific term “means for” or “step for” is used within a given claim, it is not intended that the claim be interpreted under 35 U.S.C. § 112, paragraph 6.

Claims

1. A method for grounding an electromagnetic interference (EMI) shield comprising: assembling a module comprising a module substrate, the module substrate protruding on at least one side of the module, wherein the protruding portion of the module substrate comprises one or more grounding pads; and coupling an EMI shield to the module, wherein the EMI shield comprises at least one shield component.
2. The method of claim 1, further comprising: applying an adhesive to the module for adhering the EMI shield to the module; and curing the adhesive.
3. The method of claim 1, wherein the module substrate protrudes on one side of the module, and wherein the protruding portion of the module substrate comprises two or more discrete grounding pads.
4. The method of claim 1, wherein the EMI shield comprises discrete cutouts for preventing jetted solder from expanding at unconstraint and resulting in cracking.
5. The method of claim 3, further comprising applying jetted solder to the two or more grounding pads of the module substrate.
6. The method of claim 1, wherein the module substrate protrudes on one side of the module, and

wherein the protruding portion of the module substrate comprises one elongated grounding pad.
7. The method of claim 6, further comprising: applying conductive adhesive to the one elongated grounding pad to create a conductive connection between the module substrate and the EMI shield; and curing the conductive adhesive.
8. The method of claim 6, further comprising: applying solder paste to the one elongated grounding pad to create a conductive connection between the module substrate and the EMI shield; and reflowing the solder paste.
9. The method of claim 1, wherein the module substrate protrudes on all sides of the perimeter of the module, and wherein the protruding portions of the module substrate comprise a perimeter grounding pad.
10. The method of claim 9, further comprising: applying conductive adhesive to the perimeter grounding pad to create a conductive connection between the module substrate and the EMI shield; and curing the conductive adhesive.
11. The method of claim 9, further comprising: applying solder paste to the perimeter grounding pad to create a conductive connection between the module substrate and the EMI shield; and reflowing the solder paste.
12. The method of claim 1, wherein the module substrate comprises one or more grounding pads on an underside of the module substrate.
13. The method of claim 12, further comprising: applying conductive adhesive to the one or more grounding pads on the underside of the module substrate; coupling a bottom EMI shield component to the module via the one or more grounding pads; curing the conductive adhesive; and coupling a top EMI shield component to the bottom EMI shield component.
14. The method of claim 13, wherein the coupling the top EMI shield component to the bottom EMI shield component comprises using mechanical coupling.
15. The method of claim 1, wherein the module substrate comprises grounded conductive vias which span across the thickness of the module substrate and wherein the conductive vias surround the perimeter of the module substrate.
16. The method of claim 1, further comprising: applying masking to areas determined to remain free of coating; spraying the EMI shield with a conductive coating such that the conductive coating grounds the EMI shield to the one or more grounding pads; removing the masking; and curing the conductive coating.
17. The method of claim 1, further comprising using a darkened EMI shield, wherein the darkened EMI shield is comprised of at least one of: nickel-phosphorous; oxidized nickel; graphene; or dark-painted material.
18. A system comprising: an optical module comprising: one or more optical components; an optical module substrate, wherein the optical module substrate protrudes on at least one side of the module, wherein the protruding portion of the module substrate comprises one or more grounding pads; and an EMI shield, wherein the EMI shield comprises at least one shield component.
19. The system of claim 18, wherein the EMI shield is coupled to the optical module substrate via at least one of: conductive adhesive; solder paste; or jetted solder.
20. The system of claim 18, wherein the optical module substrate comprises grounded conductive vias which span across the thickness of the optical module substrate and wherein the conductive vias surround the perimeter of the optical module substrate.